



QWETRUM TECHNOLOGIES

HR Department — Internship Program 2026

INTERNSHIP TRACK: DATA ANALYSIS

WEEK 3 — INTERNSHIP TASKS

Duration: 1 Week

Submission Day: Every Saturday by
11:59 PM

Tasks This Week: 2

TASK 01 — SQL for Data Analysis

Task 1 — SQL for Data Analysis

Use SQL to extract insights from a relational database — a core data analyst skill.

Requirements:

- Set up SQLite (or use an online SQL playground like SQLiteOnline)
- Import a dataset with at least 2 related tables (e.g., Orders and Customers)
- Write queries: JOIN, GROUP BY with aggregate functions (SUM, AVG, COUNT)
- Write a query using a subquery and one using a window function (RANK or ROW_NUMBER)
- Answer 5 business questions purely using SQL queries — show query + result for each
- Write a 150-word reflection: how is SQL different from doing this analysis in Pandas?

■ **Deliverable:** SQL file with all queries + screenshots of results + reflection document, zipped.

TASK 02 — A/B Testing Analysis

Task 2 — A/B Testing Analysis

Analyze the results of an A/B test and determine statistical significance.

Requirements:

- Create or find a dataset simulating an A/B test (e.g., two website versions and conversion rates)
- Calculate conversion rate for Group A and Group B
- Perform a Chi-Square test or Z-test to check if the difference is statistically significant
- State your null hypothesis, alternative hypothesis, and p-value clearly
- Visualize the results with a bar chart comparing both groups with confidence intervals
- Write a 200-word business recommendation: should the company switch to version B?

■ **Deliverable:** Jupyter Notebook with test data, statistical test, visualization, and recommendation.

■ **Important Instructions for All Interns**

- **No AI Tools Allowed:** Do NOT use ChatGPT, Claude, Copilot, Gemini, or any AI assistant. The purpose of this internship is YOUR learning. If you want to actually grow, do the work yourself.
- **Task Submission:** Create a ZIP file of your completed task. Upload it to your personal Google Drive folder (link shared by HR). Submit every **Saturday by 11:59 PM**.
- **LinkedIn Progress Update:** Post your weekly progress on LinkedIn. Mention what you built and what you learned. Tag **Qwetrum Technologies** in your post. This is mandatory.
- **Quality Matters:** A clean, working, well-commented submission is far more valuable than a rushed one. Take your time and do it properly.